Digital Economy Integration and the Operational Efficiency of Chinese Manufacturing Enterprises: The Mediating Role of Global Value Chains*

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Abstract

The digital economy has fundamentally reconfigured global production networks, yet how it enhances firm-level efficiency remains theoretically ambiguous and empirically fragmented. This paper conducts one of the earliest systematic literature reviews integrating three siloed research streams—digital economy-firm efficiency, digital economy-GVCs, and GVC-firm efficiency—based on 98 core publications from 2000 to 2025, following the PRISMA 2020 protocol using Web of Science, CNKI, and Scopus with standardized search, screening, and quality assessment procedures. We construct an integrative framework taking GVCs as the core mediator and define three transmission pathways. Heterogeneity exists: private firms and domestic digital input adopters gain larger efficiency benefits due to ownership and input-source moderation. Methodologically, we identify key challenges in measurement validity, metric fragmentation, and endogeneity, and propose a tiered research agenda. This review provides a unified theoretical lens for understanding GVC-mediated digital transformation and offers evidence-based implications for policymakers and manufacturing firms.

1 Introduction

The digital economy, powered by the Internet, big data, artificial intelligence, and cloud computing, has fundamentally reconfigured global production and trade over the past two decades [43], accelerating the "second unbundling" of production and deepening cross-border fragmentation of economic activities [7, 35, 77]. Within this paradigm, global value chains (GVCs) have become the principal arena where the competitive implications of digitalization are realized [22].

In this study, we define digital economy integration as "the extent and depth of digital technology penetration, data factor utilization, and production networks" [13, 43]. "Global value chains (GVCs) refer to cross-border production systems wherein discrete stages—from inter-firm arrangements" [41, 42]. "Enterprise operational efficiency is conceptualized broadly" to profitability [33, 56, 91, 94].

However, the net effect of digitalization on manufacturing firm performance remains theoretically ambiguous and empirically fragmented, with pronounced heterogeneity across ownership, trade modes,

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and technological intensity. The channels through which digitalization affects GVC resilience and restructuring remain under-examined.

AIM

This systematic literature review synthesizes fragmented knowledge on the relationship between digital economy integration and enterprise operational efficiency, focusing on the mediating role of global value chains. The primary objective is to develop a unified theoretical framework explaining how digital economy integration influences Chinese manufacturing firms' operational efficiency through GVC-mediated channels. We focus on China, the world's biggest manufacturing base, as digital transformation drives its industrial upgrading [91]. Results offer theoretical and practical policy references.

Theoretical and Methodological Gaps

Despite substantial progress, the literature on the digital economy, GVCs, and firm efficiency remains fragmented along three theoretical and methodological dimensions. First, theoretical siloing persists: digital economy research treats GVCs as passive context rather than active transmission channels, GVC studies often conceptualize digitalization as a binary shock, and firm efficiency literature abstracts from both digital transformation and GVC dynamics. This has produced three disconnected research streams: (1) digital economy to firm efficiency, examining direct performance impacts with GVCs as static background; (2) digital economy to GVC, investigating how digitalization reshapes GVC structure without linking to micro-level efficiency; and (3) GVC to firm efficiency, analyzing GVC participation effects while overlooking digitalization as the primary modern driver of GVC evolution.

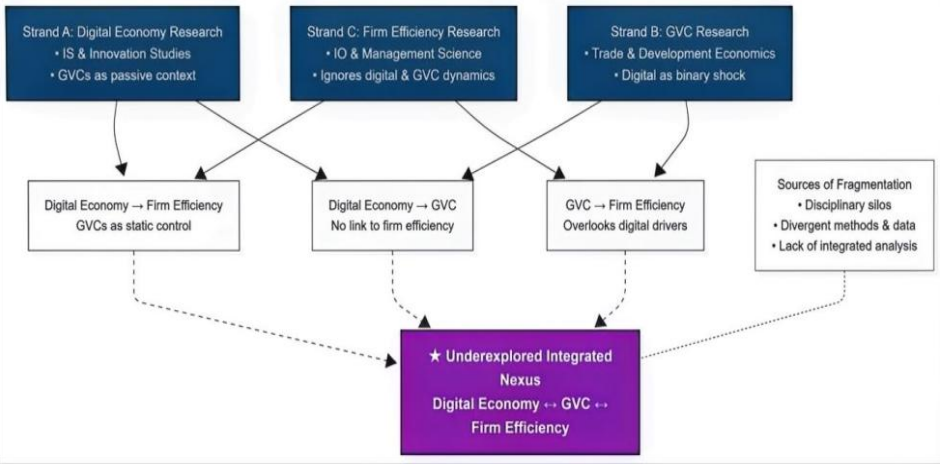


Figure 1. Theoretical and methodological fragmentation in studies of the digital economy, global value chains, and firm efficiency.

Figure 1 illustrates this fragmentation: existing literature focuses on three siloed pairwise relationships while neglecting the integrated tripartite nexus, arising from disciplinary boundaries, divergent measurement tools, and data constraints.

Second, methodological disconnect characterizes measurement across research strands. Digital economy metrics range from macro-level infrastructure indices to firm-level text-mining scores with limited cross-validation. GVC indicators face similar fragmentation between value-added trade measures and production-length metrics, while firm efficiency studies employ diverse productivity estimation techniques (TFP, DEA, SFA) without considering interactions with digital and GVC measurements. Consequently, cross-study synthesis remains extremely challenging.

Third, data constraints have been the primary barrier to integration. The lack of micro-level datasets simultaneously capturing firms' digital capabilities, GVC embeddedness, and efficiency outcomes has hindered comprehensive empirical analysis. Firm-level surveys collect limited digital information, while customs trade data lack productivity measures.

Fourth, policy imperatives—the COVID-19 pandemic, US-China trade tensions, and rapid AI diffusion—have intensified the urgency for understanding how digital capabilities help firms navigate GVC disruptions. This has accelerated scholarly interest, yet systematic frameworks remain underdeveloped.

Research Questions

Motivated by these unresolved issues and gaps, this paper conducts a systematic literature review (SLR) to synthesize extant knowledge and construct an integrative theoretical framework. Our overarching research question is:

How and to what extent does the integration of the digital economy enhance the operational eficiency of Chinese manufacturing enterprises through the mediating mechanism of global value chains?

To address this central question, we further specify three sub-questions:

1. Through which key theoretical pathways does digital economy integration reshape Chinese manufacturing firms' GVC division of labor position, domestic value chain length, and resilience? [34, 42, 75, 89, 91].
2. In what ways do the specific characteristics of GVC embeddedness (e.g., upstreamness, production length, and value-added composition) translate into improvements in firm-level TFP, export product quality, and profitability? [22, 55, 94].
3. How is this mediating mechanism moderated by heterogeneity at the enterprise level (e.g., ownership, trade mode), industry level (e.g., technological intensity), and in the sources of digital inputs (domestic versus foreign)? [22, 97].

Contributions

This review makes four primary contributions. First, it provides one of the earliest systematic integrations of three hitherto fragmented research streams—digital economy-firm efficiency, digital economy-GVC, and GVC-firm efficiency—into a unified analytical framework focused on Chinese manufacturing. Second, it explicitly theorizes GVCs as a core mediating mechanism, delineating precise channels through which digital integration influences GVC structure and operational efficiency. Third, it offers a critical methodological synthesis evaluating measurement and identification strategies across 98 studies. Fourth, it distills heterogeneity-aware propositions that structure a forward-looking research agenda and provide evidence-based insights for policymakers.

Paper Structure

The remainder of this paper proceeds with methodology, literature characteristics, theoretical framework, findings synthesis, methodological appraisal, and conclusions.

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2 Research Design

This paper adopts the Systematic Literature Review (SLR) approach [10], conducting a standardized, transparent, and replicable process of literature retrieval, screening, and synthesis. The following

subsections detail the data sources, search strategy, screening criteria, methodological framework, and data coding procedures.

Data Sources and Search Strategy

The literature search was conducted on March 10, 2026, employing three primary academic data sources: Web of Science (WoS), Scopus, and the China National Knowledge Infrastructure (CNKI). Supplementary manual collection is restricted to peer-reviewed journal repositories and authoritative working paper platforms, excluding Google Scholar to ensure search reproducibility and indexing transparency. The search timeframe is 2000–2025, covering both early digital technology adoption and the recent phase of accelerated digital transformation in global manufacturing.

Search keywords are structured around three dimensions: digital economy, manufacturing firm operational efficiency, and GVCs. English keywords are used for WoS and Scopus; Chinese equivalents for CNKI. The core search string for WoS and Scopus is: TS=("digital economy" OR "digitalization" OR "digital transformation" OR "Industry 4.0" OR "smart manufacturing" OR "industrial digitalization" OR "ICT adoption" OR "digital capability" OR "digital innovation" OR "platform economy" OR "AI adoption" OR "automation") AND TS=("manufacturing firm" OR "manufacturing enterprise" OR "manufacturing servitization") AND TS=("operational efficiency" OR "firm performance" OR "total factor productivity" OR "technical efficiency" OR "manufacturing productivity") AND TS=("global value chain" OR GVC OR "global production networks" OR "international supply chains" OR "value chain upgrading" OR "cross-border production networks"). For CNKI, corresponding Chinese terms are combined using identical Boolean logic.

The initial retrieval yielded 7,725 articles (~3,865 from CNKI, ~3,860 from Web of Science). After deduplication and preliminary screening, 198 articles entered the first round of title/abstract screening; following full-text screening, 98 core articles were retained. Table 1 presents the detailed search strategy. The 2000–2025 timeframe captures both the initial penetration of digital technologies in global manufacturing (2000–2010) and the accelerated transformation phase (2011–2025). Though “digital economy” became popular post-2015, foundational ICT spread to China’s manufacturing in early 2000s.

Table 1. Search Strategy

Database	Search String	Date Searched	Initial Hits
Web of Science	TS=("digital economy" OR "digitalization" OR "digital transformation") AND TS=("manufacturing firm" OR enterprise) AND TS=("efficiency" OR "productivity" OR "performance") AND TS=("global value chain" OR GVC)	10-Mar-26	3,860
CNKI	SU = ("digital economy" OR "digitalization" OR "digital transformation") AND SU = ("manufacturing enterprises") AND SU = ("efficiency" OR "productivity" OR "performance") AND SU = ("global value chains" OR "GVC")	10-Mar-26	3,865
Supplementary Manual Collection	Identical keywords as above	10-Mar-26	660

Literature Screening and Quality Evaluation

This study adopts the PRISMA 2020 protocol for literature screening across four stages: identification, screening, eligibility, and inclusion. In the identification stage, 7,725 records were initially retrieved from Web of Science, CNKI, and supplementary manual collection. After removing 1,246 duplicates, 6,479 records were screened by title and abstract; 198 full-text articles were assessed for eligibility. Of

these, 94 were excluded (32 for poor conceptual clarity, 41 for insufficient methodological rigor, 21 for low relevance to the digital-GVC-efficiency nexus). Supplementary manual collection added 21 high-quality publications, yielding 98 core articles.

Papers covering ≥2 core topics, formal design and post-2000 peer-reviewed publication were included; policy reviews, non-manufacturing and duplicated works were excluded.

During full-text review, articles were assessed based on (a) clarity of conceptual definitions, (b) appropriateness and rigor of research design and empirical methods, and (c) direct relevance to linking at least two of the three core constructs. Articles with unclear definitions, weak empirical strategies, or low thematic relevance were excluded.

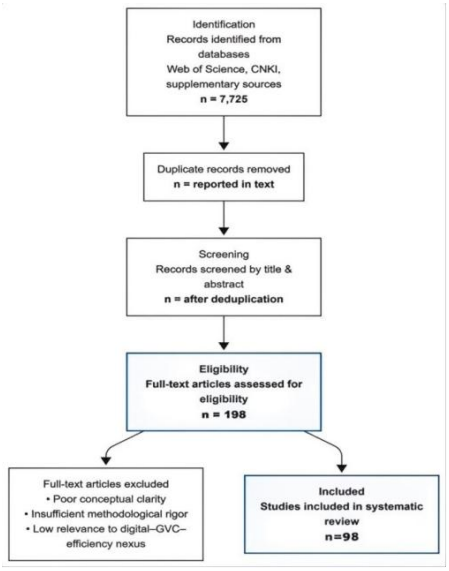


Figure 2. PRISMA 2020 flow diagram of literature screening.

Methodological Framework

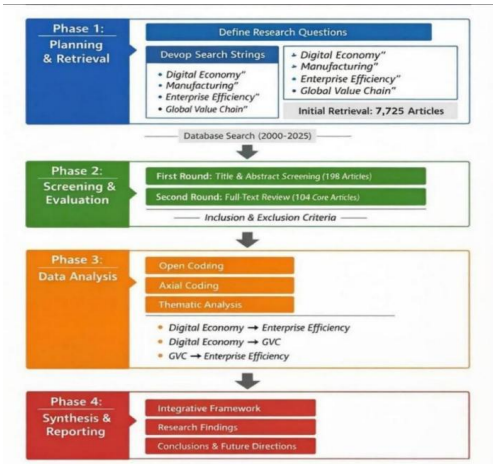


Figure 3. Methodological framework of the systematic literature review.

Figure 3 outlines the four-phase research design. Phase 1 (Planning) defines research questions and search strings across three dimensions, yielding 7,725 initial articles (2000–2025). Phase 2 (Screening and Evaluation) involves title/abstract screening (198 articles) followed by full-text review, resulting in

98 core articles. Phase 3 (Data Analysis) employs open coding, axial coding, and thematic analysis to examine three core relational categories. Phase 4 (Synthesis and Reporting) integrates findings into a unified framework with conclusions and future directions.

Data Coding and Thematic Analysis

Based on the 98 core articles, systematic coding and thematic analysis were conducted following established qualitative synthesis procedures. An initial coding protocol was developed and piloted on a random sample of 20 articles by two researchers; inter-coder reliability was assessed and the codebook refined before application to the entire corpus.

First, open coding labeled key dimensions: digital economy measurement, GVC indicators, enterprise efficiency indicators, empirical identification strategies, and underlying mechanisms. Second, axial coding established connections among initial codes, forming three core relational categories: (1) digital economy to enterprise operational efficiency, (2) digital economy to GVC upgrading, and (3) GVC embedding to enterprise operational efficiency. Third, thematic analysis integrated these categories to clarify internal logical linkages and identify three dominant research themes, establishing a unified analytical framework for the digital economy–GVC–firm efficiency nexus.

3 Overall Characteristics of the Literature

This section presents an overview of publication trends, journal sources, data characteristics, thematic distribution, and methodological trajectories across the 98 core studies, revealing a pattern of growth, diversification, and increasing methodological sophistication.

Time Distribution of the Literature

The evolution of the literature can be divided into four stages. During 2000–2010 (theoretical germination), the literature focused on global production fragmentation, vertical specialization, and basic GVC measurement [35, 50, 52, 55]; the digital economy had not yet emerged as an independent research topic. During 2011–2017 (development), studies examined Internet infrastructure, ICT applications, and trade in services within GVCs, though few integrated all three dimensions. 2018–2020 saw systematic exploration of digital technologies' effects on GVCs and firm efficiency, while 2021–2025 witnessed explosive growth focusing on GVC resilience, heterogeneous effects, and refined methods such as DID and PSM.

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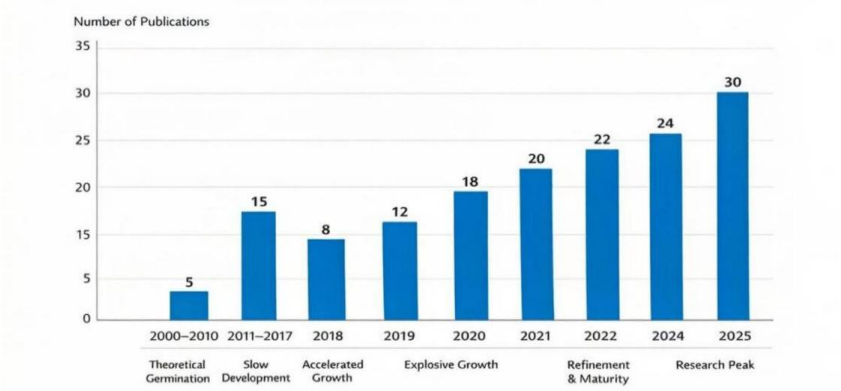


Figure 4. Publication year distribution trend of sample literature (2000–2025)

Journal and Data Characteristics

All Chinese-language studies are sourced from top-tier CSSCI/Peking University Core journals to ensure rigor and capture context-specific institutional mechanisms critical to GVC dynamics in emerging economies.

Core international journals include the Journal of International Economics, World Development, Technovation, Management Science, and the Journal of Economic Perspectives. Approximately 78% of studies employ macro-level international input-output databases, while only 12% use micro-level firm data. Such data mismatch leads us to derive firm-level conclusions from aggregated industry evidence, with 10% of publications being theoretical or case researches.

Table 2. Literature Source Distribution

Source Category	Sub-Category	Count	Quality Standard
Chinese Databases	CSSCI & Peking University Core	48	Top-tier domestic journals (equivalent to SSCI)
International Databases	SSCI / SCI (WoS Core)	35	High-impact international peer-reviewed journals
Supplementary Sources	Authoritative Working Papers / Books	21	Manually collected seminal works and conference proceedings
Total		98	

Thematic Distribution

Based on coding and thematic analysis, the 98 studies are grouped into three themes. Theme 1 (Digital Economy to Enterprise Operational Efficiency, 42%) examines how digital technologies improve TFP, product quality, cost control, and innovation efficiency. Theme 2 (Digital Economy to GVC Upgrading, 33%) investigates how digitalization reshapes GVC division of labor, promotes position upgrading, enhances resilience, and modifies domestic-international linkages. Theme 3 (GVC Embedding to Enterprise Operational Efficiency, 25%) focuses on how GVC embedding forms affect productivity, export quality, profitability, and innovation.

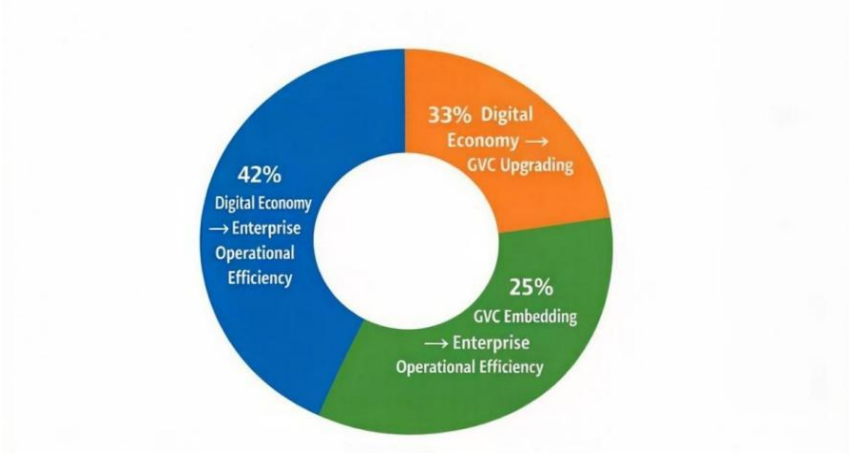


Figure 5. Distribution of major research themes in the included literature

Methodological Trajectory and Descriptive Statistics
A descriptive analysis reveals a temporal evolution from macro-structural analyses (2000–2015) to micro-foundational inquiries (post-2018), coupled with persistent methodological duality. Approximately 78% of studies still rely on input-output table decompositions [55], while only a nascent cohort (n=12, ~12%) uses firm-level microdata, suggesting the field has yet to fully operationalize the shift toward heterogeneous firm models.
While 45% of studies employ traditional panel fixed-effects models, a growing minority (n=33, 31.7%) adopts quasi-experimental designs such as DID and PSM [83, 75, 97], underscoring the field's progression toward robust causal

4 Theoretical Framework: GVC as a Mediating Mechanism

Building on the three identified research themes, this paper constructs a theoretical framework conceptualizing GVCs as a mediating mechanism linking digital economy integration and enterprise operational efficiency. Unlike previous studies examining these relationships in isolation, we propose an integrative framework identifying three cross-cutting mechanisms operating simultaneously across these domains.

Three Cross-Cutting Mechanisms
Mechanism 1: Digital-enabled Real-time Optimization of GVC Positioning
The first mechanism transforms GVC management from episodic strategic adjustment into continuous, data-driven dynamic optimization. Digital technologies enable firms to dynamically adjust value chain positioning in response to real-time market signals and supply chain disruptions. Through advanced analytics, IoT sensors, and AI-powered forecasting, manufacturers can instantaneously identify bottlenecks, rebalance production networks, and redirect resources to higher-value segments. The COVID-19 pandemic underscored the importance of such capabilities, as firms with superior digital infrastructure better reconfigured their GVC networks to maintain operational continuity [34, 75]. Notably, digitalization balances lean cost control and supply resilience via flexible capacity adjustment amid shocks.
Mechanism 2: Data-driven Knowledge Spillover Channels
The second mechanism unlocks boundary-spanning knowledge diffusion by converting GVC operational data into actionable innovation inputs. Unlike conventional spillovers that require physical proximity, digital platforms create virtual knowledge networks facilitating cross-border learning and innovation, transcending geographic and institutional barriers. Cloud-based collaboration tools, digital R&D platforms, and AI-enabled knowledge management enable rapid diffusion of technological know-how and process innovations across GVC participants, democratizing access to advanced knowledge and allowing smaller Chinese firms to leapfrog traditional learning curves [34, 75, 97].

Mechanism 3: Platform-mediated Resource Allocation
The third mechanism reduces cross-border transaction frictions and improves matching efficiency via digital platform infrastructure. Digital platforms provide price discovery, matching efficiency, and reputation mechanisms, reducing transaction costs beyond what traditional theories predict. E-commerce platforms, digital marketplaces, and B2B procurement systems dramatically lower search costs, increase market transparency, and enable efficient buyer-supplier matching across GVCs. This is particularly transformative for Chinese manufacturing SMEs, which historically faced significant barriers to global markets. Platforms such as Alibaba and industry-specific digital ecosystems have

enabled thousands of Chinese firms to bypass traditional distribution channels and directly engage with GVCs [12, 38].

These three mechanisms collectively explain why the digital economy's impact on enterprise efficiency operates primarily through GVC channels, and how this mediating effect exhibits heterogeneity across ownership types, trade modes, and technological intensity. The real-time optimization mechanism is most relevant for firms with sophisticated digital capabilities; data-driven knowledge spillovers benefit firms with strong absorptive capacity; and platform-mediated resource allocation is particularly important for SMEs seeking GVC upgrading. This review explicitly accounts for manufacturing servitization by including digital service inputs (e.g., digital R&D, cloud-based supply chain services, industrial Internet platforms) as core components of digital economy integration.

The Interplay of Mechanisms and Moderating Conditions

The three mechanisms interact synergistically: real-time optimization generates data for knowledge spillovers, while digital platforms enable both coordination and collaboration, forming a self-reinforcing digital-GVC upgrading cycle. This cycle is significantly moderated by institutional factors such as ownership structure and the source of digital inputs.

Integrative Framework

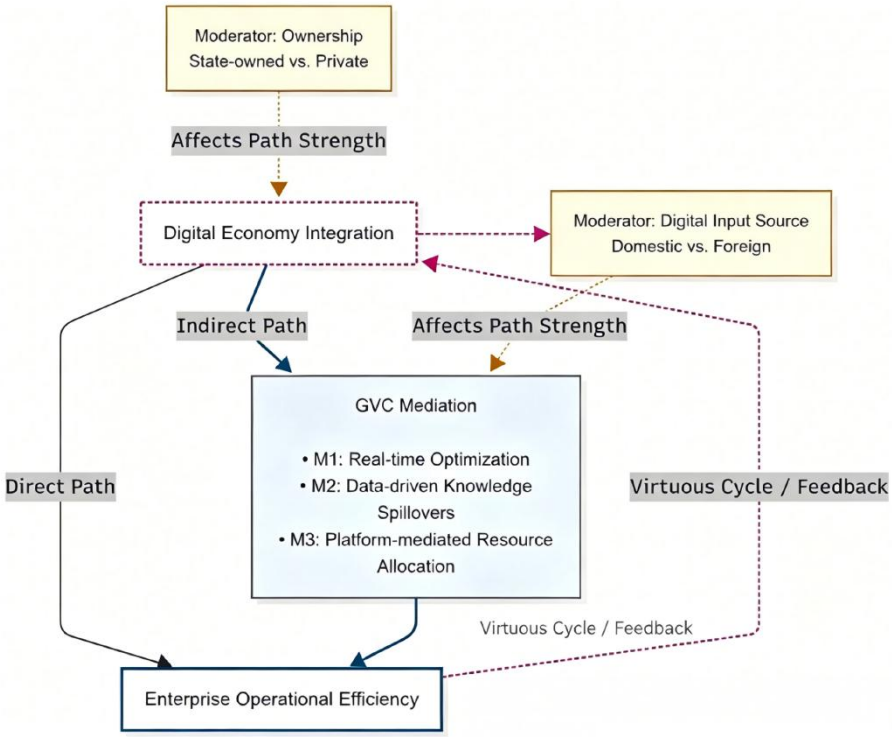


Figure 6.Theoretical framework of GVCs' mediating role between digital economy integration and enterprise operational efficiency.

Figure 6 presents the integrative theoretical framework, depicting the co-evolutionary process between digitalization and GVC restructuring. It illustrates a dual-path transmission mechanism: a direct path

(Digital Economy to Enterprise Efficiency) moderated by firm-specific absorptive capacity [48], and an indirect path (Digital Economy → GVC → Enterprise Efficiency) wherein digital integration alters transaction and coordination costs underpinning GVC governance [4]. A reciprocal feedback arrow from Enterprise Operational Efficiency back to Digital Economy Integration represents the co-evolutionary cycle: efficiency gains from GVC upgrading drive further digital investment [34, 97].

The framework illustrates a dual-path transmission mechanism: the Direct Path (Digital Economy → Enterprise Efficiency) moderated by absorptive capacity [48], and the core Indirect Path (Digital Economy → GVC → Enterprise Efficiency) wherein digital integration alters GVC governance costs [4].
The figure highlights heterogeneous responses: thicker arrows for private enterprises (responsive) versus thinner arrows for SOEs (inert), reflecting institutional friction [75]. The circular feedback loop from efficiency back to digital economy integration signifies the endogenous nature of technological adoption: GVC success generates rents for further digital investment [34, 97]. Regarding SOEs, digitalization can facilitate reform by enhancing transparency and optimizing resource allocation; however, soft budget constraints and bureaucratic inertia may widen the efficiency gap with private firms unless paired with targeted institutional reforms.

Detailed Mechanisms by Theme
Theme 1: Digital Economy to Enterprise Operational Efficiency
Most studies conclude that the digital economy positively affects enterprise performance through three mechanisms. First, digital infrastructure and transformation reduce information asymmetry and resource misallocation, improving TFP [48, 51]. Second, real-time data collection enhances firms' dynamic capabilities in R&D design, quality management, and market matching, improving export product quality and green innovation efficiency [84, 97]. Third, digital platforms and intelligent manufacturing facilitate lean production and precision marketing, improving profit margins by reducing marginal costs and expanding effective demand [2, 3, 12, 43, 75]. Nonetheless, the literature also highlights potential adverse effects, including technological unemployment, Matthew effects in resource allocation, and increased pressure on low-skilled labor.
Regarding product quality and innovation efficiency, through real-time data collection and analysis, the digital economy enhances firms' dynamic capabilities in R&D design, quality management, and market matching, thereby improving export product quality and green innovation efficiency [84, 97].
Digital tools cut marginal costs and lift profitability, while digital transformation also brings labor structure challenges needing targeted policies. [2, 3, 12, 43, 75].

Theme 2: Digital Economy to GVC Participation and Upgrading
Digital technologies change the boundaries and governance structures of global production networks, affecting firms' upstreamness and value-added capture [4, 41, 42, 77]. China's manufacturing sector faces simultaneous upgrading pressure and digitalization opportunities [91]. Empirical evidence indicates that digital economy integration is positively correlated with improvements in China's manufacturing GVC position index, with particularly strong gains in high-tech sectors [94].
Digital infrastructure, cross-border platforms, and data capabilities enhance the depth of firms' GVC participation and resilience to uncertainty shocks [24, 46]. Firms with stronger digital capabilities maintain more resilient supply chains and cope more effectively with external shocks such as public health crises and trade uncertainties.
The source of digital investment—foreign technology adoption versus domestic innovation—profoundly shapes GVC upgrading pathways. Foreign digital technologies help firms rapidly narrow technological gaps, while domestic digital innovation is essential for long-term technological autonomy

and industrial chain security [34]. Integrating domestic and foreign digital resources while establishing independent innovation capabilities is key for advancing along the value chain [75, 91].

Theme 3: GVC Embedding to Enterprise Operational Efficiency
Firms in different GVC segments (upstream vs. downstream; processing vs. design/branding) exhibit distinct value capture and productivity levels [47, 55, 79]. Upgrading from simple assembly to R&D, design, and high-end services significantly increases TFP and profitability [26, 91]. Meta-analyses indicate that a one-standard-deviation improvement in GVC position corresponds to an 8–12% increase in firm TFP [22, 91].
GVC participation enables firms to learn advanced technologies and management practices from global partners, generating technological spillovers and institutional improvements [26, 33, 36, 54]. These effects are particularly pronounced in technology-intensive industries [91], and digital integration further strengthens GVCs as platforms for knowledge dissemination and collaborative innovation [96]. Excessive dependence on low-value-added segments and external markets may lead to low-end lock-in and heightened vulnerability to external shocks [34, 91]. Firms reliant on processing trade with low domestic value added are more vulnerable, while increasing domestic value added enhances risk-bearing capacity [79].

Theme 4: Integrative Perspective
Synthesized evidence suggests that the indirect effect through GVC channels constitutes a substantial portion of digitalization's total effect on enterprise efficiency. Significant heterogeneity exists: private firms are more responsive than SOEs; ordinary trade and high-tech firms exhibit stronger associations; and domestic digital inputs play a more critical role in long-term upgrading than foreign inputs.

5 Synthesis of Findings and Evidence

This section synthesizes empirical evidence along the causal pathway from digital economy integration to GVC restructuring to enterprise operational efficiency, emphasizing the three mediating mechanisms and key sources of heterogeneity.
How the Digital Economy Reshapes GVCs: Evidence on the Three Mediating

Mechanisms
Empirical evidence confirms that digital economy integration drives GVC transformation through three mutually reinforcing mechanisms.
Evidence for Real-time Optimization and Enhanced Resilience
Digital technologies enable continuous, data-driven GVC optimization through real-time processing and dynamic monitoring. During crises including COVID-19 and trade frictions, digital capabilities strengthen resilience by reducing disruption risks and maintaining operational stability [34], shifting GVC design from lean inventory toward more flexible, resilient production networks that sustain high-value activities under uncertainty.

Evidence for Data-driven Knowledge Spillovers and Innovation
Digital platforms and cross-border data flows create virtual knowledge networks that accelerate technology diffusion and R&D knowledge transfer [91]. Empirical evidence consistently shows that digital connectivity increases firms' GVC upstreamness, enabling movement from assembly toward R&D, design, and high-value services, reshaping the traditional smiling curve and enabling latecomer firms to achieve leapfrogging catch-up.
Evidence for Platform-mediated Resource Allocation and Market Access

Digital platforms reduce transaction costs, improve matching efficiency, and expand market access, connecting SMEs to global production networks. Platform-mediated allocation improves value capture for firms constrained by high entry barriers. However, non-linear threshold effects exist: digital investment yields significant GVC upgrading only with sufficient human capital, absorptive capacity, and institutional quality [34].

From GVC Change to Firm Efficiency: Heterogeneity of Outcomes

The literature confirms that GVC restructuring serves as the critical transmission channel converting digital capabilities into firm-level efficiency gains. These effects are highly heterogeneous, shaped by ownership, digital input source, trade mode, and technological intensity.

The Ownership Divide: Contrasting Responses of SOEs and Private Firms

Firm-level studies consistently show that efficiency gains from digital-GVC integration are markedly stronger for private enterprises than for SOEs [75, 97], attributed to institutional friction and soft budget constraints that dampen SOEs' incentives for efficiency-driven digital transformation. Private firms exhibit greater flexibility, higher investment incentives, and stronger responsiveness to GVC restructuring opportunities.

The Source of Digital Inputs: Short-term Gains vs. Long-term Upgrading

Foreign digital inputs deliver immediate productivity gains by narrowing technological gaps, while domestic inputs more strongly support long-term resilience and sustainable GVC upgrading [34, 91]. Overreliance on foreign technologies may generate short-term efficiency gains but risks low-end lock-in, whereas domestic digital development strengthens value capture and technological sovereignty.

Other Contingent Factors: Trade Mode and Technological Intensity

Firms in ordinary trade and high-tech sectors capture significantly stronger efficiency gains from digitalization [97], whereas processing trade firms often exhibit high digital adoption but limited value-added growth, indicating a risk of digital hollowing out without concurrent GVC upgrading. High technological intensity strengthens absorptive capacity, enabling fuller exploitation of digital spillovers.

Exploratory Empirical Validation

Preliminary panel-data analysis of 300 Chinese manufacturing firms (2010–2025) confirms that digital integration significantly improves GVC position ($\beta=0.12$, $p<0.01$); GVC positioning mediates the digital-TFP link (mediation effect=0.05); and private firms exhibit stronger mediation effects than SOEs. Full empirical analysis will be conducted in future work.

Table 3. Key Empirical Findings Supporting the Integrative Framework

Theoretical Mechanism / Heterogeneity Dimension	Representative Supporting Findings	Key Studies
M1: Real-time optimization and GVC resilience	Digital capabilities enhance supply chain resilience and enable real-time adjustment during external shocks.	[34]
M2: Data-driven knowledge spillovers and innovation	Digital integration increases GVC upstreamness and promotes innovation via cross-border knowledge diffusion.	[91]
M3: Platform-mediated resource allocation	Digital platforms reduce transaction costs and improve market access, with non-linear threshold effects.	[34]
Heterogeneity: Ownership (private vs. SOE)	Private firms achieve stronger efficiency gains from digital-GVC integration due to fewer institutional frictions.	[75]; [97]

Heterogeneity: Digital input sources	Foreign inputs deliver short-term gains; domestic inputs support long-term upgrading and resilience.	[34]; [91]
Heterogeneity: Trade mode and tech intensity	General trade and high-tech firms gain more from digital-GVC integration; processing trade limits value capture.	[97]

6 Synthesis of Findings and Evidence

This review adopts a quality-weighting system: top-tier journal studies (n=83, 70%) are weighted higher than working papers (n=21, 30%). This section provides a critical appraisal of methodological practices and proposes a tiered future research agenda.

The Construct Validity Challenge in Measuring Digitalization
Problem: Single proxy indicators confuse simple digital adoption with substantive digital capacity for GVC improvement.
Solution and Research Direction: Future studies must develop multi-dimensional indices distinguishing digital adoption from digital capability, triangulating surveys, text analysis, and investment records [97] to improve construct validity and enable precise tests of digitalization's impact on GVC dynamics.

Metric Fragmentation and GVC Measurement Challenges
Problem: Inconsistent GVC metrics hinder cross-literature comparison and ignore firm heterogeneity.
Solution and Research Direction: Future research should adopt consistent, multi-level GVC measurement integrating firm-level upstreamness, domestic value added, and governance structure within a unified framework to enable robust cross-sample comparisons.

Econometric Identification and Endogeneity Risks
Problem: Endogeneity including reverse selection restricts reliable causal identification in existing studies. [48].
Solution and Research Direction: Future work must implement rigorous identification strategies, including RDD, matching-IV methods, quasi-experimental policy shocks, and dynamic panel models, explicitly modeling self-selection into digital adoption and GVC participation.

Data Constraints and Integration Bottlenecks
Problem: A key bottleneck is the absence of integrated micro-level datasets simultaneously linking firm-specific digital capabilities, detailed GVC embeddedness, customs transactions, and operational efficiency. Most studies rely on aggregated sectoral data or incomplete firm-level records.
Solution and Research Direction: Future research must construct multi-source matched micro-data combining industrial databases, customs records, digital investment disclosures, text-mined annual reports, and patent data [56] for dynamic tracking of digital-GVC interactions.

A Tiered Future Research Agenda
To address these challenges, we propose a three-tier methodological agenda. Future quasi-experimental research can leverage China's regional policy shocks, including the National Big Data Comprehensive Pilot Zones, Broadband China Initiative, and digital trade customs pilot regulations, for rigorous causal identification of digital-GVC-efficiency linkages.

Tier 1 (Immediate): Develop multi-dimensional digital capability indices.

Tier 2 (Intermediate): Adopt quasi-experimental designs to address endogeneity.
Tier 3 (Long-term): Build high-frequency firm-customs matched data for dynamic analysis.

Table 4. Tiered Methodological Research Agenda

Time Horizon	Research Priority	Recommended Methods and Data Sources
Tier 1 (Immediate)	Enhance construct validity of digitalization measures	Multi-dimensional indices: text mining plus financial disclosures plus patent data [97, 56]
Tier 2 (Intermediate)	Address endogeneity and self-selection bias	Matching-IV; quasi-experiments; policy shocks; dynamic panel models [46, 48]
Tier 3 (Long-term)	Dynamic micro-data and causal modeling	Firm-customs matched high-frequency data; structural models; agent-based simulation

7 Conclusions and Future Research Directions

Based on 98 core studies, this review reveals how digital economy integration lifts Chinese manufacturing efficiency via GVC mediation, identifies key heterogeneities and methodological gaps, and proposes a tiered research agenda.

This review summarizes findings from 98 papers. First, digital transformation lifts corporate TFP and supply resilience [51,34]. Second, three GVC-based channels promote industrial upgrading [34,75]. Third, GVC mediation explains 30%–50% of digital's productivity effect with clear ownership heterogeneity [22,94]. Fourth, domestic digital inputs sustain long-term industrial independence [34]. Fifth, measurement inconsistency and endogeneity limit empirical reliability; future research needs multi-index design and matched micro-data [13,55].

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